1. 2013/2014

Question 1 [8 Marks]

$$\frac{14!}{2!\,4!\,3!} = 302702400$$

i) How many ways are there to order the letters of the word INDESCREETNESS? (2 marks)

ii) A computer scientist is trying to discover the keyword for a financial account. If the keyword consists only of 10 lower case characters (e.g., 10 characters from among the set: a, b, c, ..., y, z) and no character can be repeated, how many different unique arrangements of characters exist? (1 marks)

26

$^{26}P_{10} = 1.928 \times 10^{13}$

iii) There are 12 hospitals in a town. How many different ways can 7 patients are sent to the hospitals so that no 2 patients may be in the same hospital. (1 marks)

$^{12}P_7 = 3991680$

iv) 12 identical chairs must be arranged in the following manner:

- The number of students per row has to be at least 3.
- Number of row has to be at least 2.
- Equal number of students has to be seated in a row.

How many different arrangements are possible? (4 marks)

$$^{12}C_6 \times {^6C_6} + {^{12}C_4} \times {^8C_4} \times {^4C_4} + {^{12}C_3} \times {^9C_3} \times {^6C_3} \times {^3C_3} = 405174$$

(i) 302702400 (ii) 1.928×10^{13} (iii) 3991680 (iv) 533520

i) $\dfrac{14!}{2!\,4!\,3!}$

$= \boxed{302702400}$ #

ii) $^{26}P_{10} = \dfrac{26!}{(26-10)!} = \dfrac{26!}{16!} = \boxed{1.928 \times 10^{13}}$ #

iii) $^{12}P_7 = \dfrac{12!}{(12-7)!} = \dfrac{12!}{5!} = \boxed{3991680}$ #

iv) $^{12}C_6 \times {^6C_6} + {^{12}C_4} \times {^8C_4} \times {^4C_4} + {^{12}C_3} \times {^9C_3} \times {^6C_3} \times {^3C_3}$

* $= \boxed{405174}$

** @ $\dfrac{12!}{6!\,6!} + \dfrac{12!}{4!\,4!\,4!} + \dfrac{12!}{3!\,3!\,3!\,3!} = \boxed{405174}$ *

2. 2013/2014

Question 2 [12 Marks]

i) A bagel shop has 3 union bagels, 3 poppy seed bagels, 3 egg bagels, 4 salty bagels and 4 pumpernickel bagels. How many way are there to choose

a) A dozen bagels (1 marks)

b) Seven bagels of egg bagel, pumpernickel bagels and salty bagels with three egg bagels and no more than two salty bagels? (4 marks)

ii) A linear algebra class consists of 10 mathematics majors and 12 computer science majors. A team of 12 has to be selected from this class. Find the number of ways of selecting a team if

a) The team has 6 from each discipline (2 marks)

b) The team has a majority of computer science majors (5 marks)

(i)	(a)	6188	(b)	52
(ii)	(a)	194040	(b)	332904

i) a) $^{17}C_{12} = \dfrac{17!}{(17-12)!\,12!} = \dfrac{17!}{5!\,12!} = \boxed{6188}$

b) 3 Egg + 2 Salty + 2 Pump @

3 Egg + 1 Salty + 3 Pump @

* 一定要有 Egg/Salty/Pump，所以 Salty 不可以 0

$^3C_3 \times {^4C_2} \times {^4C_2} + {^3C_3} \times {^4C_1} \times {^4C_3} = \boxed{52}$

ii) a) $^{10}C_6 \times {^{12}C_6} = \boxed{194040}$

b) $^{12}C_7 \times {^{10}C_5} + {^{12}C_8} \times {^{10}C_4} + {^{12}C_9} \times {^{10}C_3}$

$+ {^{12}C_{10}} \times {^{10}C_2} + {^{12}C_{11}} \times {^{10}C_1} + {^{12}C_{12}} \times {^{10}C_0}$

* $= \underline{330024+1} = 333025$

3. 2014/2015

Question 1 **[10 Marks]**

A father, mother, 2 boys, and 3 girls are asked to line up for a photograph. Determine the number of ways they can line up if

a) there are no restrictions (1/2 marks)

b) the parents stand together (3 marks)

c) the parents do not stand together (1/2 marks)

d) all the females stand together (3 marks)

e) all the males cannot stand together (3 marks)

(a) 7! (b) 1440 (c) 3600 (d) 576 (e) 1440

a) $7! = 5040$

b) $2!\,6! = 1440$

c) $5! \times {}^6P_2 = 3600 \leftarrow \downarrow \_ \downarrow \_ \downarrow \_ \downarrow \_ \downarrow \_ \downarrow$

d) $4! \times 4! = 576 \leftarrow \boxed{M + 3G}, F, 2$ Boys

e)

$\downarrow \_ \downarrow \_ \downarrow \_ \downarrow \_ \downarrow$

$4! \times {}^5P_3 = 1440$

Mother + 3girls

F + 2 boys

4. 2014/2015

Question 2 **[15 Marks]**

a) There are 12 different-colored cubes in a bag. How many ways can Randall draw a set of 4 cubes from the bag? Determine whether the problem represents a permutation of combination.

(2 marks)

b) Suppose there are 15 girls and 18 boys in a class. In how many ways can 2 girls and 2 boys be selected for a group project?

(5 marks)

c) A pizza menu allows you to select 4 toppings at no extra charge from a list of 9 possible toppings. In how many ways can you select 4 or fewer toppings?

(8 marks)

(a) 495 (b) 16065 (c) 256

a) ${}^{12}C_4 = 495$ (combination)

b) ${}^{15}C_2 \times {}^{18}C_2 = 16065$

c) ${}^{9}C_4 + {}^{9}C_3 + {}^{9}C_2 + {}^{9}C_1 + {}^{9}C_0$

$= 256$

5. 2015/2016

QUESTION 1 [7 Marks]

(i) Given 3 sets of integers. $A = \{1,3,5\}, B = \{4,6\}, C = \{0,2,7,9\}$. How many ways are there to choose one integer from set A, B, or C? (1 mark)

(ii) A Thai restaurant offers 5 fish dishes, 3 meat dishes, 3 vegetables dishes and 2 rice dishes for dinner service.

a) How many different dinner meals of 5 dishes are possible if any dish can be chosen only once? (1 mark)

b) How many different dinner meals are possible if each meal can consists of 1 fish dish, 1 meat dish, 1 vegetable dish and 1 rice dish. (1 mark)

c) How many different dinner meals are possible if each meal can consists of 2 fish dishes, 2 meat dishes and 1 vegetable dish but each dish can be chosen only once? (1 mark)

d) How many different dinner meals of four dishes are possible if the meal must consists only 1 fish dish, 1 meat dish, 1 vegetable dish and 1 rice dish or the meals can consists only 1 fish dish, 1 meat dish, and 2 different vegetable dishes? (3 marks)

(i) 9 ways

(ii) (a) 1287 ways (b) 90 ways (c) 90 ways (d) 135 ways

i) ${}^3C_1 + {}^2C_1 + {}^4C_1 = 3 + 2 + 4$
$= 9$

ii) a) ${}^{13}C_5 = 1287$

b) ${}^5C_1 \times {}^3C_1 \times {}^3C_1 \times {}^2C_1 = 90$

c) ${}^5C_2 \times {}^3C_2 \times {}^3C_1 = 90$

d) ${}^5C_1 \times {}^3C_1 \times {}^3C_1 \times {}^2C_1 + {}^5C_1 \times {}^3C_1 \times {}^3C_2$
$= 135$

6. 2015/2016

QUESTION 2 [18 Marks]

(i) The university attire code provides guideline for top attire, bottom attire and footwear as shown in Table 1.

Table 1: University dress code

Dress Code	Male	Female
Top wear	(i) collared T-shirt (ii) long sleeved shirt (iii) neck tie (iv) national outfit (v) uniform shirt (vi) vest 6	(i) blouse (ii) long sleeve shirt (iii) collared T-shirt (iv) national outfit (v) uniform shirt (vi) vest 6
Bottom wear	(i) slacks (ii) track bottom (iii) national outfit (iv) uniform pants 4	(i) slacks (ii) track bottom (iii) national outfit (iv) uniform pants (v) long skirt 5
Foot wear	(i) leather shoes (ii) sports shoes (iii) sandals (iv) covered shoes 4	(i) leather shoes (ii) sports shoes (iii) sandals (iv) covered shoes 4
Head covering (Optional)	Religion wear 1	(i) square scarf (ii) shawl (iii) mini telekung 3

a) How many different complete attire (top wear, bottom wear, foot wear) can a female student wear with **or** without head covering? (4 marks)

b) How many different complete attire can a female student wear if they wear a top with a vest, together with bottom wear and foot wear? (2 marks)

c) How many different complete attire (top wear, bottom wear, foot wear) can a male student or a female student wear if male students have to wear slack and both female and male have to wear covered shoes on Monday? (5 marks)

(ii) A bracelet is made from 5 alphabet beads as shown in Figure 1.

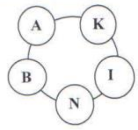


Figure 1

a) How many designs of the bracelet can be made from the 5 beads in any possible arrangement? (1 mark)

b) How many designs of the bracelet can be made if the bead with alphabet **A** and **N** should be side by side? (1 mark)

c) How many designs of the bracelet can be made if the bracelet should consists the sequence of beads **A**, **I**, **N** in any arrangement? (3 marks)

d) If the 5 beads are cut loose from the bracelet and rearranged in linear sequence, how many initials of 3 alphabet beads can be made? (2 marks)

(i) (a) 480 (b) 100 (c) 36

(ii) (a) 24 (b) 12 (c) 12 (d) 60

i) a) ${}^6C_1 \times {}^5C_1 \times {}^4C_1 \times {}^3C_1 + {}^6C_1 \times {}^5C_1 \times {}^4C_1$

$= 480$

b) $1 \times {}^5C_1 \times {}^5C_1 \times {}^4C_1$

$= 100$

c) ${}^6C_1 \times 1 \times 1 + {}^6C_1 \times {}^5C_1 \times 1$

$= 36$

ii) a) $\frac{(5-1)!}{2} = 12$

b) $\frac{(4-1)! \times 2!}{2} = 6$

c) $\frac{(3-1)!}{2} = 1$

d) $\underline{5} \times \underline{4} \times \underline{3}$

$= 60$

7. 2016/2017

QUESTION 1 **[10 MARKS]**

(a) A computer access password consists of from three to five letters chosen from the 26 in the alphabet with repetitions allowed. How many different passwords are possible?

(2 marks)

(b) Suppose there are three roads from city A to city B and five roads from city B to C.

(i) How many ways is it possible to travel from city A to C via city B? (2 marks)

(ii) How many different round-trip routes are there from city A to B to C to B and back to A? (3 marks)

(iii) How many different routes are there from city A to B to C to B and back to A in which no road is traversed twice? (3 marks)

(a) 12,355,928 ways

(b) (i) 15 ways (ii) 225 ways (iii) 120 ways

a) _ _ _

@

_ _ _ _

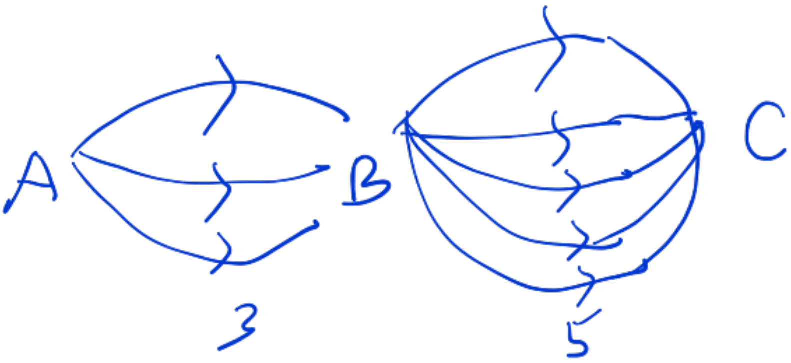
@

_ _ _ _ _

$26^3 + 26^4 + 26^5$

$= 12355928$

b)



i) $3 \times 5 = 15$

ii) $3 \times 5 \times 5 \times 3 = 225$

iii) $3 \times 5 \times 4 \times 2 = 120$

8. 2016/2017

QUESTION 2 **[15 MARKS]**

6 8 7

(a) There are six students from SCSJ, eight from SCSR and 7 from SCSV in a programming club. Find the number of orders in which the six students from this club can win the first six prizes if all students attend the competition and the winners are

(i) all members of SCSJ (2 marks)

(ii) all members of SCSR (2 marks)

(iii) all members of SCSV (2 marks)

(iv) all members of the same course. (1 mark)

(b) A father with five children – Nabilah, Syahmi, Hazirah, Haris and Basyir, takes his three children at a time to the zoological garden, as often as he can without taking the same three children together more than once. Find the number of times

(i) the father will go to the zoological garden. (2 marks)

(ii) each child will go to the zoological garden. (2 marks)

(iii) Hazirah will not go to the zoological garden. (1 mark)

(c) There are two round tables with seating capacities of five for table 1 and four people for table 2. Nine people are considered to be seated around those two tables.

(i) In how many ways people can be selected to seat around table 1? (½ mark)

(ii) In how many ways people can be arranged around table 1? (1 mark)

(iii) In how many ways people can be arranged around table 2? (1 mark)

(iv) In how many ways can nine people be seated around the two tables? (½ mark)

(a) (i) 720 (ii) 20160 (iii) 5040 (iv) 25920

(b) (i) 10 (ii) 6 (iii) 4

(c) If the students regard that all the questions are related:

(i) 126 (ii) 24 (iii) 6 (iv) 18144

If the students regard that all the questions are not related:

(i) 126 (ii) 3024 (iii) 756 (iv) 18144

a) i) $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

ii) $8 \times 7 \times 6 \times 5 \times 4 \times 3 = 20160$

iii) $7 \times 6 \times 5 \times 4 \times 3 \times 2 = 5040$

iv) $720 + 20160 + 5040 = 25920$

b) i) $1 \times {}^5C_3 = 10$

ii)

iii) $1 \times {}^4C_3 = 4$

c) i) ${}^9C_5 = 126$

ii) $(5-1)! = 24$

iii) $(4-1)! = 6$

iv) ${}^9C_5 \times (5-1)! \times {}^4C_4 \times (4-1)!$

$= 18144$

OR

c) i) ${}^9C_5 = 126$

ii) ${}^9C_5 \times (5-1)!$

$= 3024$

iii) ${}^9C_4 \times (4-1)!$

$= 756$

iv) ${}^9C_5 \times (5-1)!$

$\times {}^4C_4 \times (4-1)!$

$= 18144$

9. 2017/2018

QUESTION 1 **[15 MARKS]**

a) Three coins are tossed and the outcomes are placed in a row.

(i) How many outcomes are there? (1 mark)

(ii) How many outcomes contain at least two consecutive heads? (3 marks)

(iii) How many outcomes do not contain at least two consecutive heads? (1 mark)

(iv) How many outcomes do not contain exactly two heads? (2 marks)

Based on tree diagram

b) In a sightseeing group, there are 8 Singaporeans, 5 Indonesians and 6 Malaysians.

(i) In how many ways to select a pair of individuals of distinct nationalities from this group of tourist? $^8C_1 \times {}^5C_1 + {}^8C_1 \times {}^6C_1 + {}^5C_1 \times {}^6C_1 = 118$ (4 marks)

(ii) In how many ways to select a team of 3 tourists of distinct nationalities from this group of tourist? $^8C_1 \times {}^5C_1 \times {}^6C_1 = 240$ (2 marks)

(iii) In how many ways to select a representative for this group of tourist? (2 marks)

$8+5+6 = 19$ or $^8C_1 + {}^5C_1 + {}^6C_1 = 19$

(a)	(i)	8 ways	(ii)	3 ways	(iii)	5 ways	(iv)	5 ways		
(b)	(i)	118 ways	(ii)	240 ways	(iii)	19 ways				

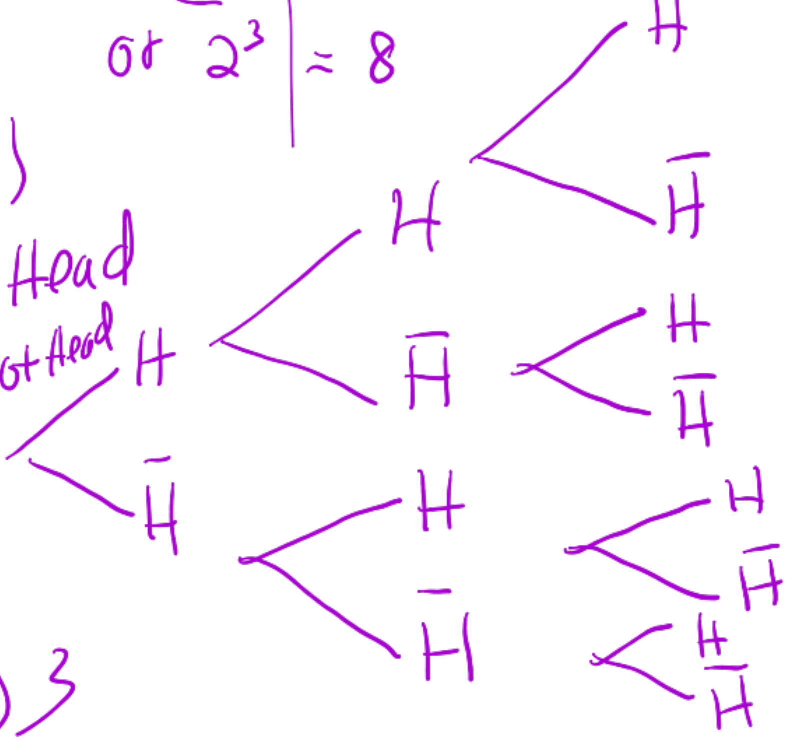
a) i) $2 \times 2 \times 2 = 8$

or $2^3 = 8$

ii)

H - Head

$\bar{H}$ - Not Head



HHH

$HH\bar{H}$

$H\bar{H}H$

$H\bar{H}\bar{H}$

$\bar{H}HH$

$\bar{H}H\bar{H}$

$\bar{H}\bar{H}H$

$\bar{H}\bar{H}\bar{H}$

ii) 3

iii) 5

iv) 5

QUESTION 2 **[10 MARKS]**

a) How many integers, greater than 999 but not greater than 4000, can be formed with the digits 0, 1, 2, 3 and 4, if repetition of digits is allowed? (2 marks)

b) There are 2 brothers among a group of 10 persons. They are going to be arranged around a circle so that there is exactly one person between the two brothers.

iii)

(i) In how many ways to select a person to be seated in between the brothers?

8

$(8-1)! = 5040$ (1 mark)

(ii) In how many ways to arranged all of them around the circle? (1 mark)

(iii) In how many ways to arrange the group around a circle so that there is exactly one person between the two brothers? (2 marks)

$(8-1)! \times (2\times1\times1) \times 8 = 80640$

(c) Suppose 7 students are staying in a hall in a hostel and they are allotted 7 beds. Among them, Arvin does not want a bed next to Anju because Anju snores. Then, in how many ways can you allot the beds to all the students? (4 marks)

(c)

(a) 376 ways

(b) (i) 8 ways (ii) 5040 ways (iii) 80640 ways

(c) 3600 ways

$5! \times {}^6P_2 = 3600$

a) $999 < x \leq 4000$

0, 1, 2, 3, 4

(1, 2, 3) 3个选择

$\underline{3} \times \underline{5} \times \underline{5} \times \underline{5} = 375$

(4) $\underline{1} \times \underline{\overset{0}{1}} \times \underline{\overset{0}{1}} \times \underline{\overset{0}{1}} = 1$

$375 + 1 = 376$

2018/2019

QUESTION 1 **[10 Marks]**

(i) Ricky is going for a winter holiday and thinking of dressing warm for the winter. He will be layering three shirts over each other, and two pairs of socks. If he has twenty shirts to choose from, along with twelve different kinds of socks, how many ways can he layer up? $^5C_4 \times {^3C_3} \times 6!$ (3 marks)

$^{20}C_3 \times {^{12}C_2} = 75240$ BYBYBYB ~~BYBYB~~

(ii) There are 4 red cups, 5 blue cups and 3 yellow cups. In how many ways can you arrange these so that each yellow cup is in between two blue cups? (4 marks)

S C S C S C S C S C   C S C S C S C S C   S S S

(iii) A company has 7 software engineers and 5 civil engineers. In how many ways can they be seated in a row so that no two of the civil engineers will sit together? (3 marks)

ii) 5 ↓ 4 ↓ 3 ↓ 2 ↓ 1   4 red

$5! \times {^4P_3} \times 5! = 345600$

iii)

↓ _ ↓ _ ↓ _ ↓ _ ↓ _ ↓ _ ↓ _ ↓

$7! \times {^8P_5} = 33868800$

software   civil

QUESTION 2 CSCSCSCSC **[15 Marks]**

(i) How many bit strings of length 8 either start with a 1 bit or end with the two bits 00?

case1 : $^{2}C_{1}$ ~~$^{12}C_{2}$~~ × ~~$^{10}C_{9}$~~ $^{8}C_{2}$ (4 marks)

case2 : $^{10}C_{2}$ × $^{10}C_{2}$

(ii) There are 12 intermediate stations between two places A and B. Find the number of ways in which a train can be made to stop at 4 of these intermediate stations so that no two stopping stations are consecutive? $^{12}C_4$ (2 marks) ^{12}C

A .... B

(iii) A cricket club in University of ABC is organizing a cricket tournament. Each team should plays one match with every other team.

2,3 3,4 4,5 5,6 6,7
2,4 3,5 4,6 5,7
2,5 3,6 4,7
2,6 3,7
2,7

(a) Find the number of matches in this tournament if there are 7 teams registered for the tournament. (2 marks)

(b) If the tournament should have 45 matches, find the number of teams that need to register in this tournament. (3 marks)

steps

Hint: $^{n}C_{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)(n-3)\ldots(1)}{r!(n-r)(n-r-1)\ldots(1)}$

(iv) In a small village, there are 25 families, of which 18 families have at most 2 children. In a rural development programme 16 families are to be chosen for assistance, of which at least 14 families must have at most 2 children. In how many ways can the choice be made? 25 (4 marks)

2018/2019

QUESTION 3

at most … 10 C 14

[12 Marks]

5 6 7

3 plays → 5 women

at least 3 women

(i) A theatre performs 7 plays in one season. Five women are each cast in 3 of the plays. Show that some play has at least 3 women in its cast. (3 marks)

< 14

(ii) There are 50 baskets of apples. Each basket contains no more than 14 apples. Show that there are at least 4 baskets containing the same number of apples. (3 marks)

(iii) A bowl contains 10 red marble and 10 blue marbles. A girl selects marbles at random without replacement.

(a) How many marbles must she select to be sure of getting 3 marbles of the same colour? (4 marks)

(b) What is the least number of marbles must she select to be sure of getting at least 3 blue marbles?

(2 marks)

2019/2020

QUESTION 1 **[10 Marks]**

(a) How many numbers between 100 and 1000 use only odd digits, no digit being repeated? (2 marks)

(b) In an examination there are three multiple choice questions and each question has 4 choices. Determine number of ways in which a student can fail to get all answer correct. (2 marks)

(c) How many non-negative integer solutions are there to the following expressions?

$$-5 < x < 5 \quad \text{or} \quad 12 < x < 100 \quad \text{or} \quad -\tfrac{2}{3} < x \leq \tfrac{2}{3}$$ (4 marks)

(d) At the dealership in Nabil's town, there are 10 red trucks, 5 blue trucks, 3 red cars, and 2 blue cars for sale. If Isaac is going to buy exactly one red vehicle, how many choices does he have? (2 marks)

a) $100 < x < 1000$

$1, 3, 5, 7, 9$

$\boxed{5 \times 4 \times 3 = 60}$

b) $4 \times 4 \times 4 = 4^3$

$1 \times 1 \times 1 = 1$

$\boxed{4^3 - 1 = 63}$

c) $-5 < x < 5$, $12 < x < 100$

$13-99$

$x = 0, 1, 2, 3, 4$ $\quad 99 - 13 + 1 = 87$

$-\frac{2}{3} < x < \frac{2}{3}$

$x = 1$

$\therefore \boxed{5 + 87 + 1 = 93}$

d) ${}^{13}C_1 = 13$

15 2019/2020

QUESTION 2 **[15 Marks]**

(a) Suppose you are organizing a panel discussion on banning vaping on campus. You need to arrange a list of participants consist of 5 administrators and 4 students. They will sit behind a table in the order listed.

(i) In how many ways can you list them if the administrators must sit together in group and the students must sit together in a group? 5! 4! (3 marks)

$2! \times 5! \times 4!$
$= 5760$

(ii) In how many ways can you list them if you must alternate students and administrators as shown in Figure 1? (1 mark)

$5 \times 4 \times 4 \times 3 \times 3 \times 2 \times 1 \times 2 \times 1 = 5! \times 4!$
$= 2880$

Administrator	Administrator	Student	Student	Administrator	Student	Student	Administrator	Administrator

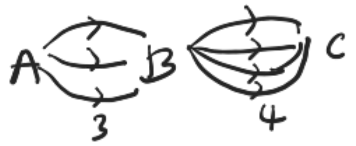
Figure 1

(b) There are 3 bridges connecting two towns, A and B. Between towns B and C there are 4 bridges. A salesperson has to travel from A to C via B. However, during his departure day, 2 of the bridges connecting town B and C are closed due to construction work for 5 days. (3 marks)

(i) Find the number of possible choices of bridges from A to C.

(ii) Find the number of possible choices of bridges for a round-trip travel A to C if the salesperson returns to town A after a week in town C.

(iii) Find the number of possible choices of bridges for a round-trip travel A to C if no bridge is repeated. The round-trip travel is done in the same day.



b) i) $3 \times 2 = 6$

ii) $3 \times 2 \times 4 \times 3$
$= 72$

iii) $3 \times 2 \times 1 \times 2$
$= 12$

(c) There are 12 objects and 12 boxes. Out of 12, 5 objects cannot fit into 3 small boxes. How many arrangements can be made, such that each object can be put in only 1 box? (3 marks)

9 Big box
3 Small box

5 object big
7 object small

c) ${}^{7}P_3 \times 9! = 76204800$
or ${}^{7}C_3 \times 3! \times {}^{9}C_9 \times 9!$
$= 76204800$

(d) In a playground 3 sisters and 8 other girls are playing together. In a particular game, how many ways can all the girls be seated in a circular order so that

(i) the 3 sisters are seated together? (3 marks)

(ii) the 3 sisters are not seated together? (2 marks)

d) i)

$(9-1)! \times 3! = 241920$

ii) $(11-1)! - 241920 = 3386880$

2019/2020

QUESTION 3 **[15 Marks]**

(a) From 5 consonants and 4 vowels, how many words (with or without meaning) can be formed using 3 consonants and 2 vowels? $^5C_3 \times {}^4C_2 \times 5!$ (3 marks)

$= 7200$ #

(b) A boy has 3 library tickets and 8 books of his interest in the library. Of these 8, he does not want to borrow Mathematics Part II, unless Mathematics Part I is also borrowed. In how many ways can he choose the three books to be borrowed? (3 marks)

$1 \times 1 \times {}^6C_1 + {}^7C_3 = 41$ #

(c) Aunt Rokiah went to the florist shop to buy a bouquet with four roses. The flower shop had white, yellow and red roses. How many different flowers bouquet can a florist make for Aunt Rokiah create? (2 marks)

$n = 3, \ r = 4$

$\frac{(n+r-1)!}{r!(n-1)!}$

$\frac{(3+4-1)!}{4!(3-1)!} = \frac{6!}{4!\,2!} = 15$ #

(d) Students in Year 3 in Faculty of Engineering can choose two different courses for the elective subjects. Their two choices must be from two different schools.

- The School of Computer Science offers 5 different courses;
- The School of Electrical Engineering offers 6 different courses;
- The School of Mechanical Engineering offers 4 different courses.

How many different ways are there of choosing two courses? (4 marks)

$^5C_1 \times {}^6C_1 + {}^5C_1 \times {}^4C_1 + {}^6C_1 \times {}^4C_1$

$= 74$ #

(e) Find the number of different triangles can be formed by joining the vertices of

(i) square (1 mark)

(ii) pentagon (5-sided polygon) (1 mark)

(iii) octagon (8-sided polygon) (1 mark)

i) $^4C_3 = 4$

ii) $^5C_3 = 10$

iii) $^8C_3 = 56$